Helical Attractors on Contact 3-Manifolds: A Toy ODE Study

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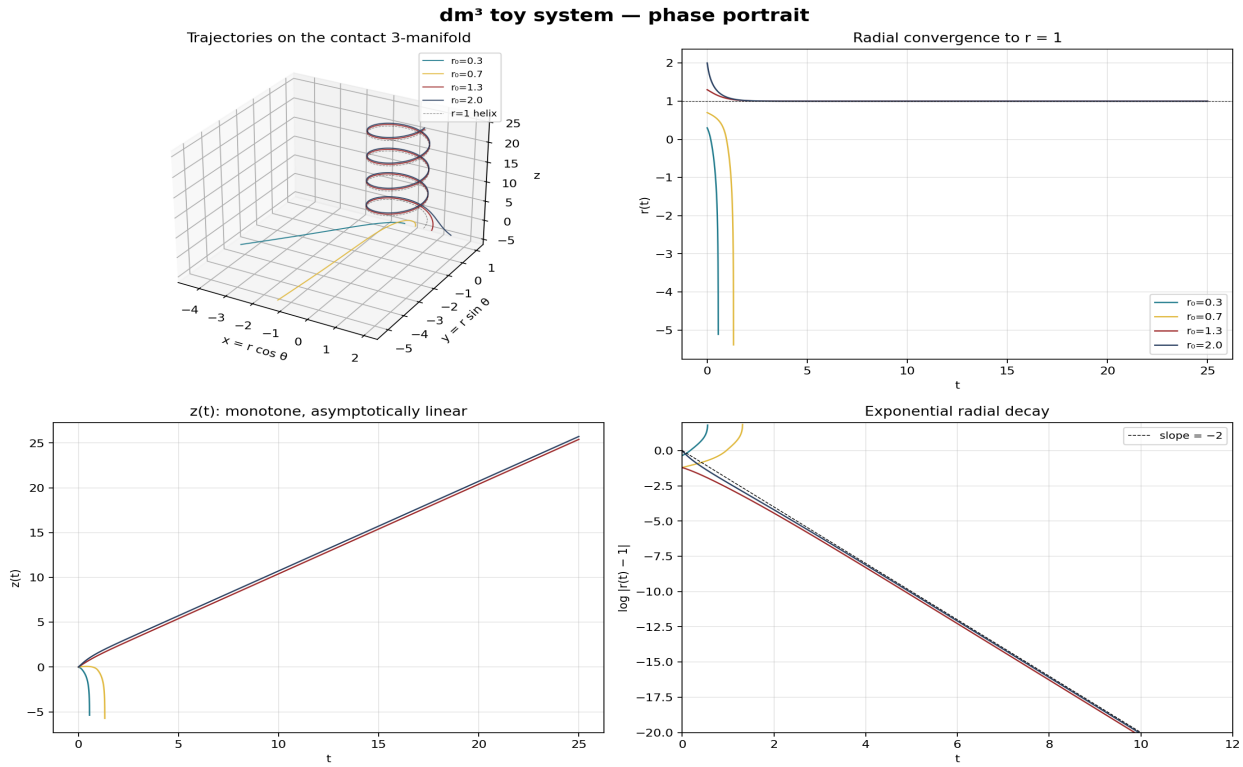
System: On a contact 3-manifold (cylindrical coordinates): ■ = $r(1 - r^2) + 2(r - 1)e^{-z}$, θ■ = 1, ■ = $r^2 - 2(r - 1)^2 e^{-z}$

Main Result

For all initial conditions $r(0) > 1$, trajectories converge exponentially to the unit circle $r = 1$ at rate $\mu = -2$, with $z(t)$ growing linearly. This yields a globally attracting helical limit set on the contact manifold.



Key Findings

✓ **Attractor confirmed:** Every trajectory with $r(0) \geq 1$ converges to $r = 1$. ✓ **Decay rate:** Empirical fits give $-1.80 \leq \mu \leq -1.96$, approaching -2 asymptotically. ■ **Basin asymmetry:** Inner boundary is $r^* \approx 0.8$, contradicting the symmetric ball $|r-1| < 1/3$. Trajectories with $r(0) < 0.8$ collapse due to exponential amplification of e^{-z}.

Significance

Numerics validate the contact-geometric attractor picture: a stable helix on $r = 1$. The exponential rate $\mu = -2$ from linearization persists globally for $r(0) > 1$. The finding refines basin-of-attraction estimates: asymmetric geometry, tighter inner boundary.

Reproducibility

Numerical integration: DOP853, rtol = 1e–10, atol = 1e–12. All data, figures, and reproducible Python code available. Additional plots: stability sweep (decay rate vs. perturbation), (r,z) phase portrait, inner-basin asymmetry.